Deep Learning Based Medical Image Analysis For Early Disease Detection

Testing Report

Phase III project submission | 30 October 2026

Student / institution: enter the approved cover details. No identity or approval has been invented.

Objectives and environment

Tests cover metadata and data loading, preprocessing, split integrity, model initialization, metrics, checkpoint save/load, end-to-end training mechanics, real dataset audit and model-backed inference. Windows, Python 3.11, PyTorch 2.15.0+cu132, torchvision 0.29.2+cu132, RTX 3050 Laptop GPU (CUDA).

Executed test cases

ID	Test scope	Expected result	Actual result	Status
T-01	Fixture metadata adapter	Validated records and labels	Passed	PASS
T-02	Grouped and predefined split	No leakage; official test preserved	Passed	PASS
T-03	Image transform	Consistent tensor shape	Passed	PASS
T-04	Model forward/checkpoint reload	Correct output shape	Passed	PASS
T-05	Metric calculation	Finite metrics and confusion matrix	Passed	PASS
T-06	One-epoch fixture pipeline	Train and checkpoint execute	Passed	PASS
T-07	ISIC source audit	Readable data, manifest and duplicates checked	1,275 records; no unreadables/missing/duplicates	PASS
T-08	GPU model prediction	Checkpoint loads and gives probabilities	ISIC held-out image predicted; probabilities returned	PASS
T-09	Grad-CAM inference	Valid overlay dimensions	224x224 map; source-sized overlay	PASS
T-10	Streamlit inference and state	Result persists and Grad-CAM renders	Verified in browser	PASS
T-11	Full pytest suite	All tests pass	13 passed in 8.12 s	PASS
T-12	Compilation / dependencies	No compile or dependency errors	compileall passed; pip check clean	PASS
T-13	NIH and APTOS full-data tests	Audit and train	Pending actual dataset access	PENDING

Method and summary

Synthetic fixtures validate software mechanics only; they are not medical evidence. The real ISIC audit, training, held-out evaluation, CUDA inference, app prediction and Grad-CAM were separately exercised. NIH/APTOS integration remains untested until the full sources are available. Final command: `.venv/Scripts/python.exe -m pytest -q --basetemp

.tmp_pytest_final` (13 passed in 8.12 s). Bytecode compilation succeeded and `pip check` reported no broken requirements.

Known outstanding deliverables: public deployment, NIH/APTOS source data, and a recorded .mov demo. A concise recording script and checklist are provided.